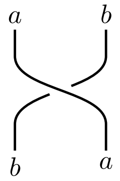
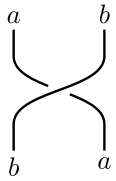
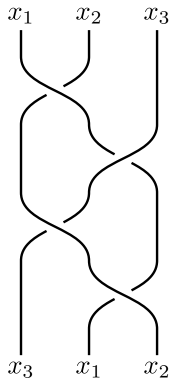
# BraidTeX 0.2 — feature showcase

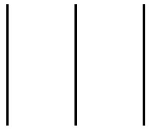
## 1. Crossings, identity, colours

Positive $\sigma_1$ (left strand over), negative $\sigma_1^{-1}$, a three-strand word, identities:

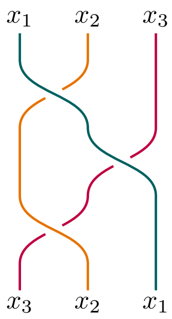


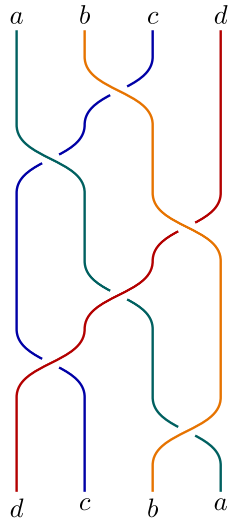






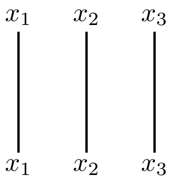
One colour per strand; colours follow strands; default palette:

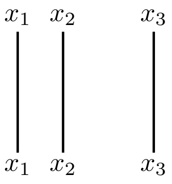


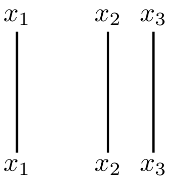


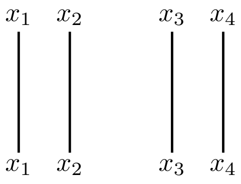
## 2. Automatic layout from parentheses

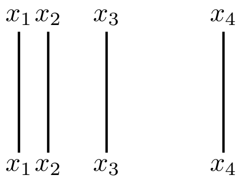
$x_1x_2x_3$, $(x_1x_2)x_3$, $x_1(x_2x_3)$, $(x_1x_2)(x_3x_4)$, $((x_1x_2)x_3)x_4$:



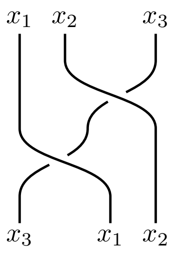


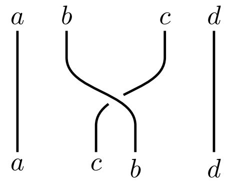




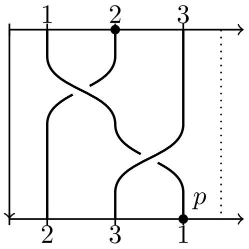


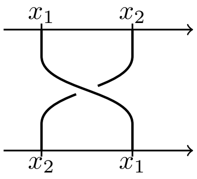
Different source and target parenthesizations:





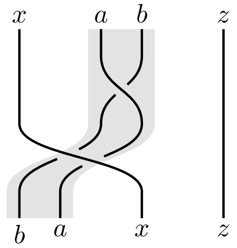
## 3. Axes and marked points

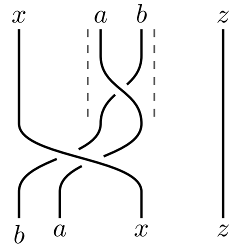


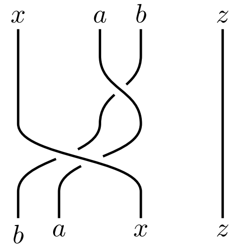


## 4. Composition: an inner braid replaces a strand

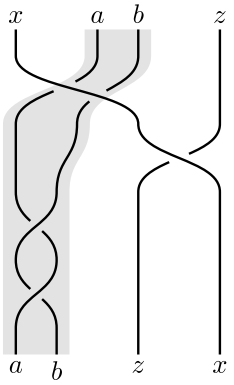
Outer braid $\sigma_1$ on $x\,y\,z$; strand $y$ is replaced by the braid $\sigma_1$ on $a\,b$. **Mode 1, shadow** (default) shades the whole region the strand is replaced in; best when the ribbon travels through crossings with other strands. **Mode 2, separator** draws two dashed lines beside the inner braid only; best when the inner braid should read as a box inside a strand.

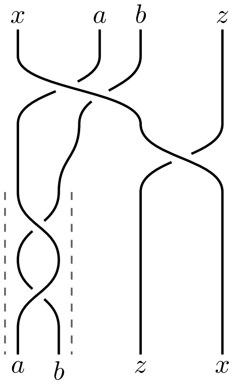


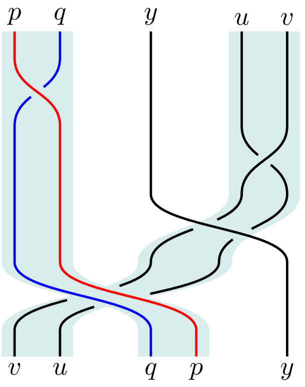




Inner braid at the bottom (`at=bottom`), shadow and separators; then two substitutions in one braid (colours, `shadowcolor`):

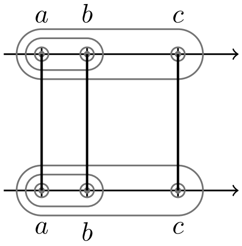


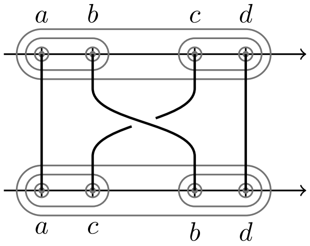


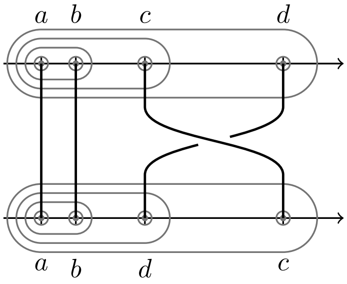


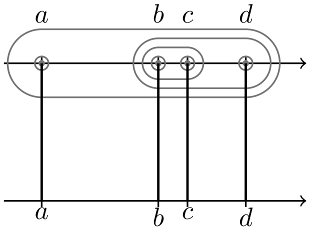
## 5. Little 2-disc boundary pictures

Each parenthesized group is one disc; $((ab)c)$ puts $a, b$ in an inner disc, which together with $c$ lies in the outer disc. Discs are drawn for the input (top) and the output (bottom) configuration.



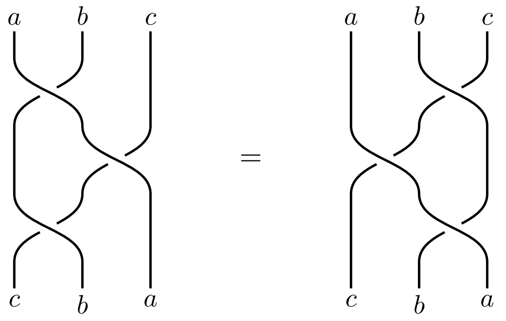


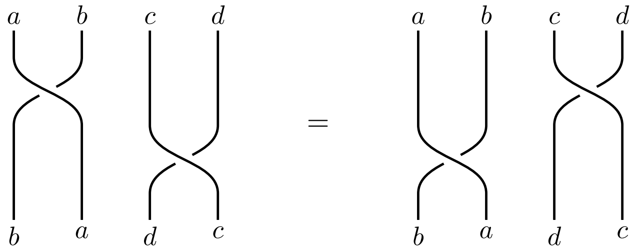


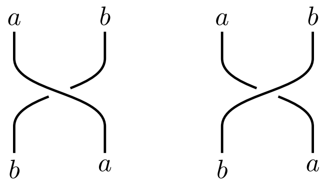


## 6. Standard identities

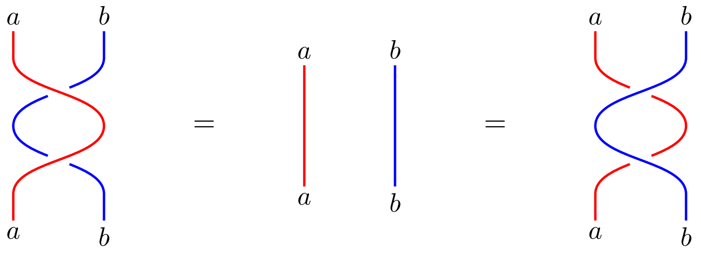
Yang–Baxter / braid relation, far commutativity, $\sigma\sigma^{-1} = 1$, the basic crossings:



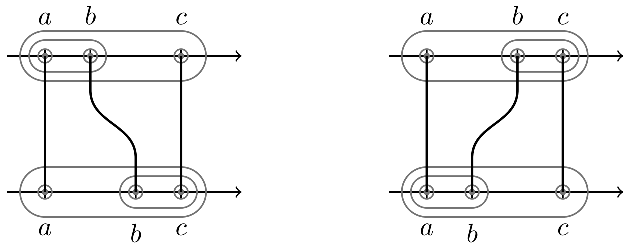




$\sigma_1$ $\sigma_1^{-1}$



The associator as a braid between different parenthesizations (and its inverse):



Pentagon and hexagon (nodes, arrows and labels are generated; all configurable):

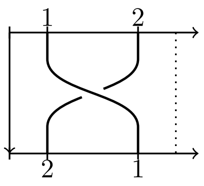
$$\begin{tikzcd}
 & & ((a \otimes b) \otimes c) \otimes d \arrow[dll, "\alpha_{a\otimes b,c,d}"'] \arrow[drr, "\alpha_{a,b,c} \otimes \mathrm{id}_d"] & & \\
(a \otimes b) \otimes (c \otimes d) \arrow[dr, "\alpha_{a,b,c\otimes d}"'] & & & & (a \otimes (b \otimes c)) \otimes d \arrow[dl, "\alpha_{a,b\otimes c,d}"] \\
 & a \otimes (b \otimes (c \otimes d)) & & a \otimes ((b \otimes c) \otimes d) \arrow[ll, "\mathrm{id}_a \otimes \alpha_{b,c,d}"] &
\end{tikzcd}$$

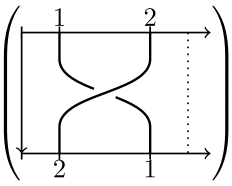
$$\begin{tikzcd}
 & A \otimes (B \otimes C) \arrow[r, "c_{A,B\otimes C}"] & (B \otimes C) \otimes A \arrow[dr, "\alpha_{B,C,A}"] & \\
(A \otimes B) \otimes C \arrow[ur, "\alpha_{A,B,C}"] \arrow[dr, "c_{A,B} \otimes \mathrm{id}_C"'] & & & B \otimes (C \otimes A) \\
 & (B \otimes A) \otimes C \arrow[r, "\alpha_{B,A,C}"'] & B \otimes (A \otimes C) \arrow[ur, "\mathrm{id}_B \otimes c_{A,C}"'] &
\end{tikzcd}$$

Restyled (own objects, no tensor sign, other associator symbol, node boxes, colour, scaled):

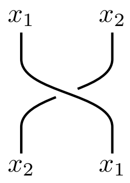
$$\begin{tikzcd}
 & & \boxed{((W\,X)\,Y)\,Z} \arrow[dll, "\Phi_{W\,X,Y,Z}"'] \arrow[drr, "\Phi_{W,X,Y} \otimes \mathrm{id}_Z"] & & \\
\boxed{(W\,X)\,(Y\,Z)} \arrow[dr, "\Phi_{W,X,Y\,Z}"'] & & & & \boxed{(W\,(X\,Y))\,Z} \arrow[dl, "\Phi_{W,X\,Y,Z}"] \\
 & \boxed{W\,(X\,(Y\,Z))} & & \boxed{W\,((X\,Y)\,Z)} \arrow[ll, "\mathrm{id}_W \otimes \Phi_{X,Y,Z}"] &
\end{tikzcd}$$

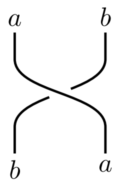
## 7. Inside expressions and inline

$\psi(\tau) =$ 

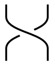
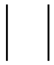
$\cdot \Bigg($ 

$\Bigg)^{\nu},$ (1)

$\sigma_1$ 

$=$ 

(2)

Inline: $\sigma_1 =$  and id $=$  sit on the math axis next to ordinary symbols.

## 8. Source scanning

```
\braid{\bin{x_1 x_2}\bcr{1}}   % shown literally, never transformed
```